Rossmoyne Senior High School

Semester One Examination, 2022

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

Section One: Calculator-free

SOLUTIONS

WA student number: In figures

In words

Your name

Number of additional answer booklets used (if applicable):

Time allowed for this section

Reading time before commencing work: five minutes
Working time: fifty minutes

Materials required/recommended for this section

To be provided by the supervisor
This Question/Answer booklet
Formula sheet

To be provided by the candidate
Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	54	35
Section Two: Calculator-assumed	12	12	100	96	65
				Total	100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.

2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.

3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.

4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.

5. It is recommended that you do not use pencil, except in diagrams.

6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.

7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section One: Calculator-free **35% (54 Marks)**

This section has **seven** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.
In general: For 2 mark questions allow answer only unless otherwise stated.
For 3 mark questions only one of the stated lines of working need to be shown unless otherwise stated.
Probabilities don't need to be simplified unless otherwise stated.
Questions 10 and 11 are units/rounding questions.

Question 1 **(6 marks)**

Solve each of the following equations.

(a) $\frac{x}{2}+\frac{2x}{3}=\frac{x+1}{2}$. (2 marks)

Solution
$\frac{x}{2}+\frac{2x}{3}=\frac{x+1}{2}$ $\frac{3x+4x}{6}=\frac{x+1}{2}$ $7x=3x+3$ $4x=3$ $x=\frac{3}{4}$
Specific behaviours
✓ simplifies to third line ✓ obtains correct solution

(b) $3x^3=12x^2$. (2 marks)

Solution
$3x^3=12x^2$ $3x^3-12x^2=0$ $3x^2(x-4)=0$ $x=0, x=4$
Specific behaviours
✓ factorises ✓ both correct solutions

(c) $(x+5)^2-49=0$. (2 marks)

Solution
$(x+5)^2-49=0$ $(x+5)^2=49$ $x+5=\pm 7$ $x=-5\pm 7$ $x=-12, x=2$
Specific behaviours
✓ arranges equation into form $a^2=b^2$ ✓ both correct solutions

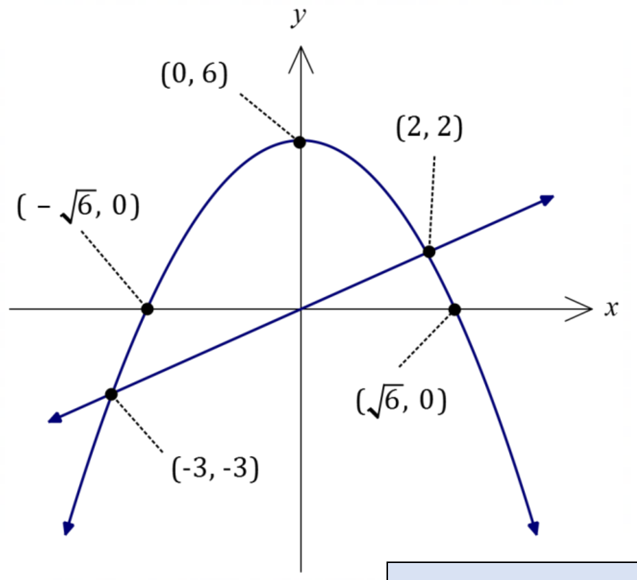
Alternate Solution
$(x+5)^2-49=0$ $x^2+10x+25-49=0$ $x^2+10x-24=0$ $(x+12)(x-2)=0$ $x=-12, x=2$
Specific behaviours
✓ simplifies to correct quadratic ✓ both correct solutions

Question 2 **(6 marks)**

(a) Solve the equation $6 - x^2 = x$. (2 marks)

Solution
$x^2 + x - 6 = 0$ $(x-2)(x+3) = 0$ $x = 2, -3$
Specific behaviours
✓ equates to zero and factorises ✓ correct solutions

(b) Sketch the graphs of $y = 6 - x^2$ and $y = x$ on the axis below, showing the coordinates of all axes intercepts of the parabola and any points of intersection of the graphs. (4 marks)



Solution
See graph
Specific behaviours
✓ symmetrical parabola, correct turning point ✓ labels roots of parabola ✓ correct straight line ✓ labels both points of intersection

Question 3 **(8 marks)**

(a) Expand $(x+4)(x-1)(x-4)$. (2 marks)

Solution
$(x+4)(x-4)(x-1) = (x^2-16)(x-1)$ $= x^3 - x^2 - 16x + 16$
Specific behaviours
✓ expands any pair of binomials ✓ correct expansion

(b) Let $f(x) = x^3 - x^2 - 14x + 24$.

(i) Calculate $f(-2) - f(2)$. (2 mark)

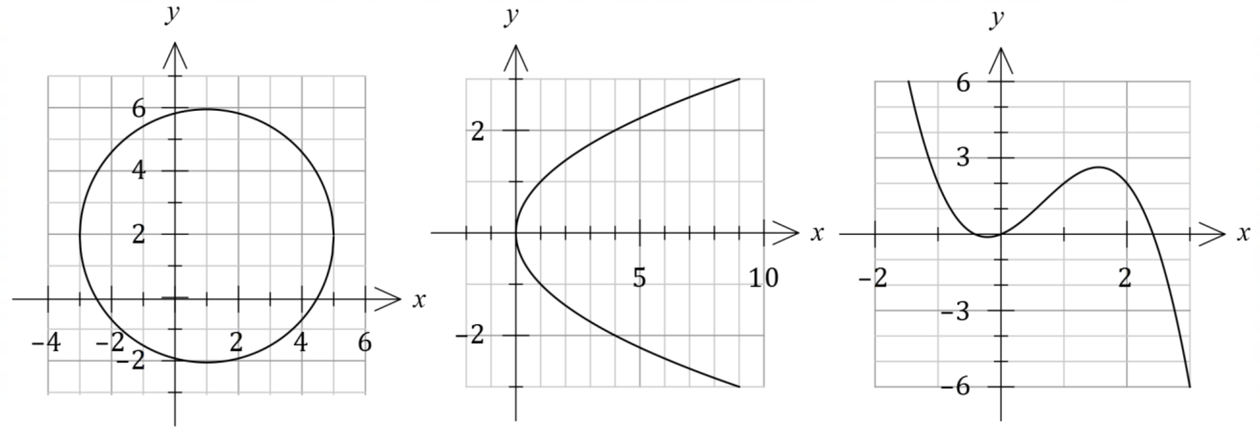
Solution
$f(-2) = -8 - (4) - 14(-2) + 24 = 52 - 12 = 40$ $f(2) = 8 - (4) - 14(2) + 24 = 32 - 32 = 0$ $f(-2) - f(2) = 40 - 0 = 40$
Specific behaviours
✓ correct value $f(2) = 0$ ✓ correct difference

(ii) Solve $f(x) = 0$. (4 marks)

Solution
Since $f(2) = 0$ then $(x-2)$ is a factor of $f(x)$. By inspection, $f(x) = x^3 - x^2 - 14x + 24$ $= (x-2)(x^2+x-12)$ $= (x-2)(x-3)(x+4)$ Hence $f(x) = 0$ when $x = -4, x = 2, x = 3$.
Specific behaviours
✓ obtains factor using (i) ✓ expresses as linear and quadratic factors ✓ factors quadratic (must have) ✓ determines correct solutions from factors

Question 4 **(8 marks)**

The graphs of the function $y = f(x)$ and two relations are shown below.



(a) Explain how the vertical line test can be used to distinguish a function from a relation. (2 marks)

Solution
The test concludes that a relation is a function if and only if no vertical line intersects the relation more than once. Otherwise, graph is simply a relation.
Specific behaviours
✓ includes reference to all possible vertical lines or to x values ✓ includes reference to no more than one intersection for function

(b) State the equation of the parabolic relationship. (1 mark)

Solution
$y^2 = x$
Specific behaviours
✓ correct equation

(c) Determine $f(3)$. (1 mark)

Solution
$f(3) = -6$
Specific behaviours
✓ correct value

(d) Solve $f(x) = 2$. (1 mark)

Solution
$x = -1, \quad x = 1, \quad x = 2$
Specific behaviours
✓ all correct solutions

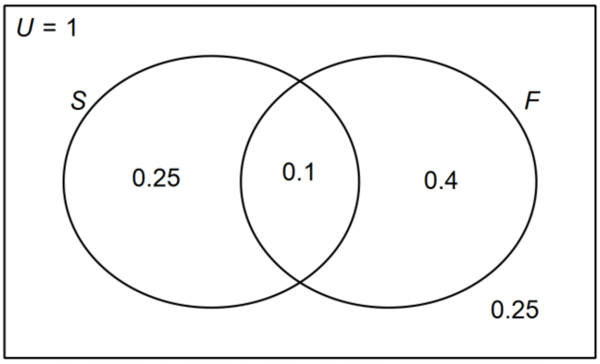
(e) The equation of the circle is $x^2 + y^2 + ax + by = c$, where a, b and c are constants. Determine the value of each constant. (3 marks)

Solution
$(x-1)^2 + (y-2)^2 = 4^2$ $x^2 - 2x + 1 + y^2 - 4y + 4 = 16$ $x^2 + y^2 - 2x - 4y = 11$ $a = -2, \quad b = -4, \quad c = 11$
Specific behaviours
✓ correct factored form of circle ✓ expands and simplifies ✓ uses expanded form to state each value

Question 5 **(5 marks)**

On a certain day, patients report at a general practice to see a doctor.
50% of the patients have a fever.
It is found that the probability of a patient having a sore throat is 0.35,
while the patient having both a fever and a sore throat is 0.1.

(a) Complete all regions of the Venn diagram to represent this situation. (2 marks)



Solution
Specific behaviours
✓ 0.1 and 0.4 correct
✓ all correct probabilities

(b) What is the probability that a patient selected at random does not have a sore throat. (1 mark)

Solution
0.65
Specific behaviours
✓ correct probability

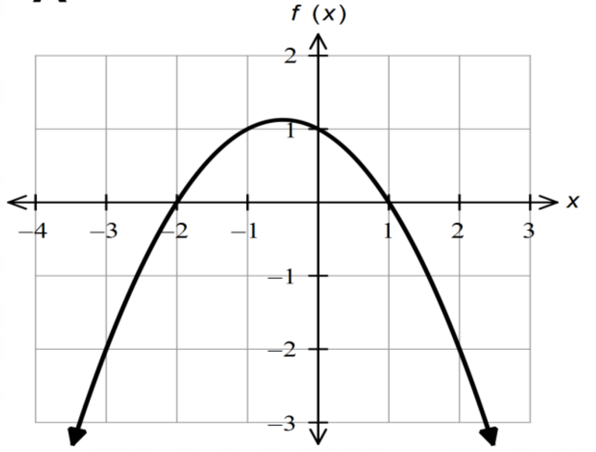
(c) If a patient selected at random has a sore throat, what is the probability that they have a fever? (2 marks)

Solution
$\frac{0.1}{0.35} = \frac{10}{35}\ or\ \frac{2}{7}$
Specific behaviours
✓ determines denominator of 0.35
✓ correct probability (no decimal in fraction)

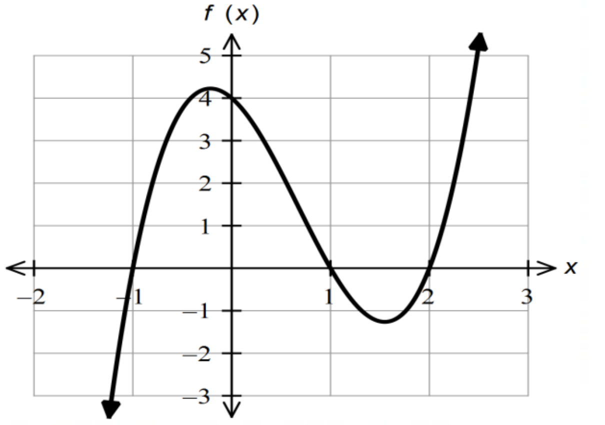
Question 6 **(7 marks)**

Consider the graphs drawn below.

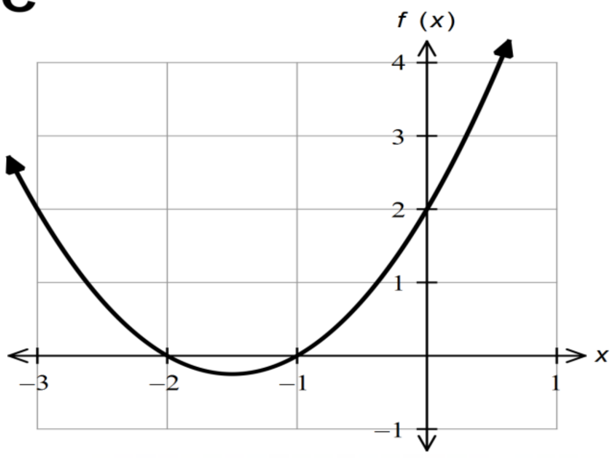
A



B



C



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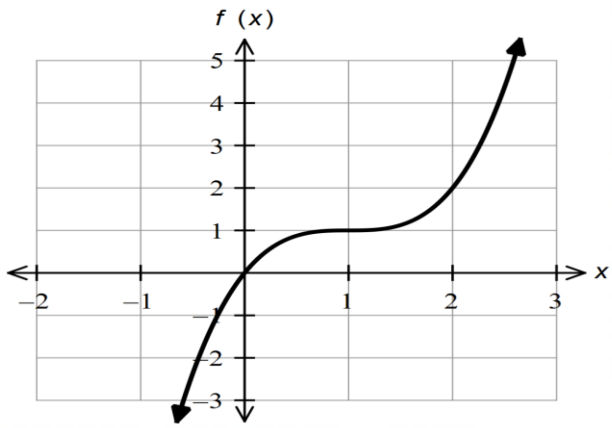
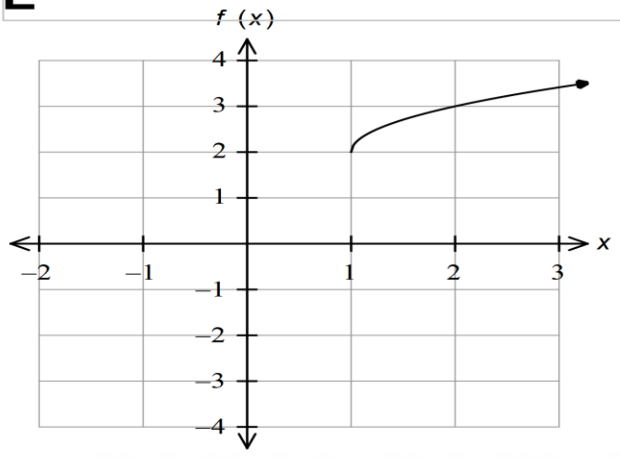
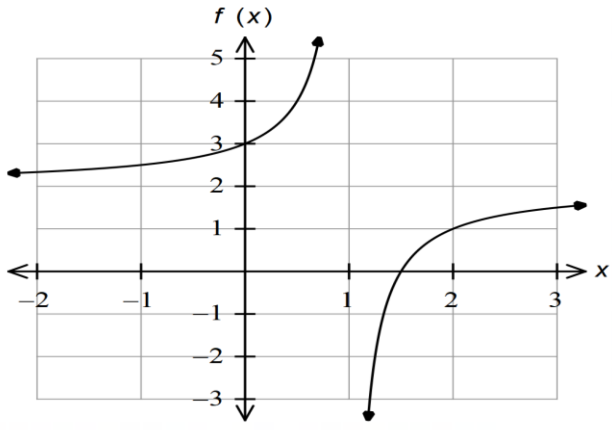


E



F



The equations of the graphs drawn are given below.

$f(x) = a(x-1)(x+2)$; $f(x) = \sqrt{(x+b)} + 2$; $f(x) = (x-1)^3 + c$

$f(x) = x^2 + dx + 2$; $f(x) = e(x^2-1)(x+f)$; $f(x) = \dfrac{1}{1-x} + g$

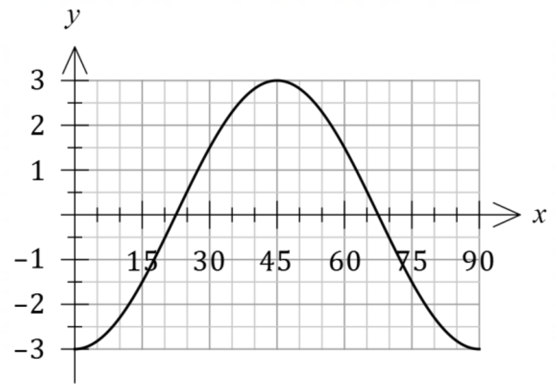
Determine the values of *a*, *b*, *c*, *d*, *e*, *f* and *g*, writing your answers in the table below. **(7 marks)**

a	b	c	d	e	f	g
-0.5	-1	1	3	2	-2	2

Solution
Specific behaviours
one mark per correct value

Question 7 **(14 marks)**

(a) The graph of $y = a\cos(bx)$ is shown. State the value of the constant a and the value of the constant b. (2 marks)



Solution
$a = -3, \qquad b = 4$
Specific behaviours
✓ correct value of a
✓ correct value of b

(b) Point P lies on the unit circle with centre O so that the anticlockwise angle measured from the positive x-axis to the line OP is θ, $0 \leq \theta \leq 2\pi$. Determine the size of θ when P has coordinates $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$. (2 marks)

Solution
$\tan\theta = -\frac{1}{2} \div \frac{\sqrt{3}}{2} = -\frac{1}{\sqrt{3}} \to \theta = \frac{11\pi}{6}$
Specific behaviours
✓ indicates method (reasoning or sketch of unit circle)
✓ correct angle

(c) Solve the equation $1 + \sin(2x - 30°) = 0$ when $0 \leq x \leq 360°$. (3 marks)

Solution
$1 + \sin(2x - 30°) = 0$ $\sin(2x - 30°) = -1$ $2x - 30° = 270°, 630°$ $2x = 300°, 660°$ $x = 150°, 330°$
Specific behaviours
✓ indicates $\sin^{-1}(-1) = 270°$
✓ one correct solution
✓ both correct solutions

(d) Determine an exact value for $\cos\frac{\pi}{12}$, simplifying fully with a rational denominator. (3marks)

Solution
$\cos\frac{\pi}{12} = \cos\left(\frac{\pi}{4} - \frac{\pi}{6}\right)$ $= \cos\frac{\pi}{4}\cos\frac{\pi}{6} + \sin\frac{\pi}{4}\sin\frac{\pi}{6}$ $= \frac{\sqrt{2}}{2} \times \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \times \frac{1}{2}$ $= \frac{\sqrt{6}+\sqrt{2}}{4} \quad or \quad \frac{\sqrt{3}\sqrt{2}+\sqrt{2}}{4} \quad or \ldots$
Specific behaviours
✓correct difference and expands using difference formula ✓ substitutes exact values ✓ simplifies correctly over denominator of 4

(e) Solve $2\sin\left(x - \frac{\pi}{4}\right) + \sqrt{2}\cos x = 1$ for $0 \leq x \leq 2\pi$. (4 marks)

Solution
$2\sin\left(x - \frac{\pi}{4}\right) + \sqrt{2}\cos x = 1$ $2\left(\sin x\cos\frac{\pi}{4} - \cos x\sin\frac{\pi}{4}\right) + \sqrt{2}\cos x = 1$ $2\left(\left(\frac{\sqrt{2}}{2}\right)\sin x - \left(\frac{\sqrt{2}}{2}\right)\cos x\right) + \sqrt{2}\cos x = 1$ $\sqrt{2}\sin x = 1$ $\sin x = 1/\sqrt{2}$ $x = \frac{\pi}{4}, \frac{3\pi}{4}$
Specific behaviours
✓ expands using difference formula ✓ substitutes exact values ✓ simplifies to $\sin x = 1/\sqrt{2}$ ✓ correct solution

End of Paper

Supplementary page

Question number: _________